

PROJECT PLAN & TIMELINE

DairyPro 360 — All-in-One Dairy Management System

Document ID	PP-DMS-001
Version	1.0
Prepared By	Project Manager Agent
References	BRD-DMS-001 v1.1 (Validated)
Date	2026-05-25
Project Duration	6 Months (26 Weeks)
Estimated Budget	INR 28–35 Lakhs
Methodology	Agile Scrum — 2-week sprints

1. Project Methodology

DairyPro 360 will follow an **Agile Scrum** methodology with 2-week sprints. Each sprint will include planning, daily standups, sprint review, and retrospective. The project is divided into 5 phases spanning 26 weeks. A dedicated CI/CD pipeline will be set up from Week 1 to enable continuous integration and deployment.

2. Team Structure & Roles

Role	Agent / Person	Responsibility
Project Manager	PM Agent	Sprint planning, risk tracking, stakeholder comms
Business Analyst	BA Agent	BRD, FRD, user stories, UAT support
Backend Developer	Dev Agent 1	REST APIs, DB schema, integrations, authentication
Frontend Developer	Dev Agent 2	React UI, dashboards, mobile responsiveness
Quality Analyst	QA Agent	Test plans, manual + automated testing, bug reports
Validation Agent	Val Agent	Document QA, hallucination checks, cross-validation
DevOps (part-time)	Dev Agent 1	CI/CD, server setup, backups, deployment

3. Project Phases

Phase 0: Project Initiation (Week 1–2)

- Finalise and sign off BRD v1.1
- Set up project repository (GitHub), Jira board, and CI/CD pipeline
- Environment setup: dev, staging, production servers
- Kickoff meeting with all agents and stakeholders
- BA Agent produces FRD draft
- Validation Agent validates FRD
- Sprint 0 planning

Phase 1: Foundation (Week 3–6 | Sprints 1–2)

- Database schema design and setup (PostgreSQL)
- Authentication module (JWT, RBAC, 2FA)
- User management module (admin panel)
- Master data setup: cattle, farmers, products, SKUs
- Basic React project scaffold with routing and theming
- Login, dashboard shell, and navigation UI
- QA: Unit tests for auth and user management

Phase 2: Core Operations (Week 7–14 | Sprints 3–6)

- Farm & Cattle Management module (full CRUD + alerts)

- Milk Collection module (fat/SNF entry, payment calc)
- Processing & Production batch management
- Inventory Management (stock in/out, reorder alerts)
- UI for all Phase 2 modules
- QA: Integration tests for collection → processing → inventory flow
- Mid-project review and stakeholder demo (Week 14)

Phase 3: Business Modules (Week 15–20 | Sprints 7–9)

- Distribution & Delivery module (route management, tracking)
- Sales, Billing & Finance module (GST invoices, A/R, A/P)
- HR & Payroll module (attendance, leave, payslips)
- Payment gateway integration (Razorpay)
- SMS/Email notification service setup
- UI for all Phase 3 modules
- QA: End-to-end testing of billing and payroll

Phase 4: Analytics & Integration (Week 21–23 | Sprint 10)

- Reports & Analytics module (KPI dashboard, trend charts)
- Demand forecasting (3-month moving average)
- Milk analyser API integration
- Data export (PDF/Excel) for all reports
- Performance optimisation (query tuning, caching)
- QA: Full regression test suite execution

Phase 5: Testing, Hardening & Go-Live (Week 24–26)

- User Acceptance Testing (UAT) with client
- Security penetration testing
- Bug fix sprints (critical and major bugs)
- User training (2-day programme)
- Data migration from legacy records
- Production deployment and go-live
- Hypercare support (2 weeks post go-live)

4. Sprint Timeline (Gantt Summary)

Each cell marked with ■ indicates the sprint is active for that module.

Module / Activity	W1-2	W3-6	W7-10	W11-14	W15-18	W19-20	W21-23	W24-26
BRD & FRD	■							
Environment Setup	■	■						
Auth & User Mgmt		■						
Farm & Cattle			■	■				
Milk Collection			■	■				
Processing/Inventory				■				
Distribution/Delivery					■	■		
Sales & Finance					■	■		
HR & Payroll						■		
Analytics/Reports							■	
Integrations							■	
QA & Testing			■	■	■	■	■	■
UAT & Go-Live								■

5. Key Milestones

Milestone	Target Week	Owner	Deliverable
M-01: BRD & FRD Signed Off	Week 2	BA + Val Agent	Validated BRD v1.1 + FRD v1.0
M-02: Environment Ready	Week 3	Dev Agent 1	Dev/Staging/Prod servers live
M-03: Auth Module Complete	Week 6	Dev Agent 1	Login, RBAC, user management
M-04: Core Ops Modules Live (Staging)	Week 14	Dev Agents 1+2	Farm, Collection, Processing, Inventory
M-05: Business Modules Live (Staging)	Week 20	Dev Agents 1+2	Distribution, Billing, HR
M-06: Full Feature Freeze	Week 23	PM Agent	All modules in staging, no new features
M-07: QA Sign-Off	Week 24	QA Agent	Zero critical bugs, test report approved
M-08: UAT Complete	Week 25	BA + Client	UAT sign-off document
M-09: Go-Live	Week 26	PM Agent	Production deployment complete

6. Resource Plan

Resource	Type	Phase 0	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5
Project Manager	FTE	100%	50%	50%	50%	50%	100%

Business Analyst	FTE	100%	50%	25%	25%	50%	100%
Backend Developer	FTE	50%	100%	100%	100%	100%	50%
Frontend Developer	FTE	25%	100%	100%	100%	75%	50%
QA Analyst	FTE	0%	25%	75%	75%	100%	100%
Validation Agent	AI	100%	0%	25%	25%	50%	50%

7. Budget Estimate

Category	Estimated Cost (INR)
Backend Development (6 months)	8,00,000
Frontend Development (6 months)	7,00,000
QA & Testing	3,50,000
Project Management & BA	4,00,000
Infrastructure (Cloud servers, 1 year)	1,80,000
Third-party Services (SMS, Email, Payment GW)	60,000
Licensing & Tools	40,000
Training & Documentation	80,000
Contingency (10%)	2,61,000
TOTAL ESTIMATED	28,71,000

8. Risk Register

Risk	Probability	Impact	Mitigation
Scope creep from client	High	High	Strict change control; any change requires PM approval + timeline review
Agent output inconsistency	Medium	High	Validation Agent reviews all AI-generated docs before use
Integration delays (analysers)	Medium	Medium	Manual CSV import as fallback; integration in Phase 4
Performance under load	Low	High	Load testing in Week 22-23; DB indexing and Redis caching
Data migration errors	Medium	High	Pilot migration with 10% of records + validation scripts
Go-live delay	Medium	Medium	Buffer week built into Phase 5; critical path monitored weekly